

Logical Circuit Simulators

The first choice of users to check logical correctness

Logic Circuit Simulators are software that are a crucial part of any hardware designing and a must before the fabrication is done. The software in this month's EFY-CD check the logical correctness of hardware at the gate level

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Wikipedia states, "Electronic circuit is a closed path or paths formed by the interconnection of electronic components through which an electric current can flow." Electronic circuits comprise analogue circuits and digital circuits.

There are various software available to simulate analogue and digital circuits. Simulators are devices or systems that mimic specific conditions or characteristics of a real processor or machine. They serve various purposes, ranging from training individuals to checking electronic circuits. Last month, we addressed some of the issues related to analogue circuit simulation. Here we will be addressing questions related to digital logic circuit simulators.

But before we move ahead, we need to know what is a digital logic circuit simulator? The answer is, it's a tool that can be used to check the logical correctness of any hardware design. It's role is critical because checking the logical correctness is a very crucial step in the hardware design cycle. Although there are various levels, like gate level and register-transfer level (RTL), at which the logical correctness can be checked, this article focuses at the software that can help designers to check the correctness at the gate level.

A digital logic simulator has its own advantages and disadvantages. However, when the logical correctness of any hardware is to be checked at gate level, a designer should look for a simulator using the following criteria:

1. Which of the gates (AND, OR, NOT or complex ones) does the software support?
2. Is the software useful for educational or professional purpose?
3. What is the cost of the software?
4. Is it an open source software?
5. Does the software vendor provide good support?

CEDAR Logic

It is an open source software and its latest version available is CEDAR Logic Simulator 1.5. The developers of the software are Matt Lewellyn, David Knierim, Ben Sprague, Joshua Lansford and Nathan Harro. It is being distributed under BSD licence and the software's code is written in C++ language.

What's special?

CEDAR Logic is an interactive software that basically focuses on teaching logic circuit designs to beginners in an interactive and simple way. Further, it has both low-level logic gates as well as high-level components. The software includes registers and a Z80 microprocessor emulator. It has 4-bit counting register, 8-bit counting register, 16-bit counting register and 4-bit shift register. There are input and output components too, such as switch, clock, 4-bit LED display, 8-bit LED display, etc. The software also includes D- and JK-flip-flops.

Users' feedback

This software is user-friendly and not much training is required to use it. You can select components by drag-

ging and dropping. There are certain shortcuts that are very helpful in designing a circuit. You can rotate the components that you have selected just by right clicking on them. The tutorial that comes with the software is informative and explains step-by-step how to use the software. CEDAR Logic incorporates a Z80 microprocessor and its description and usage tips are also mentioned in the software. Another great feature of this software is oscilloscope, here named as 'Oscope.' The Oscope lets you view the output of the constructed circuit graphically. It opens 10 different default pages for circuit designing. You may also look state of you components. The software allows you to print the created circuit and has an export feature. It can export the circuit into bitmap format, so that it can be used in wordprocessors or for photo editing.

Suggestion

A reset button in the play, pause area would be helpful to release the states immediately. It would have been really helpful if the oscilloscope could have resolution feature, which is available in other software. The software allows export of circuits only in bitmap format.
Home page: <http://cedarlogic.scienceontheweb.net/>
Download: <http://sourceforge.net/projects/cedarlogic/files/>
Platform supported: Windows

Karnaugh Map

Karnaugh Map is also an open source software available for cross-platform